1. Algebraický tvar KČ

1. Vypočtěte: [1; 19/13]

a) $i^3 + i^{13} + i^{23} + i^{33} + i^{43}$

b) $i \,.\, i^2 \,.\, i^3 \,.i^4 \,.\ldots\ldots i^{50}$

c) $i + i^2 + i^3 + i^4 + \cdots + i^{50}$

d) $1 + i^4 + i^8 + i^{12} + \cdots + i^{52}$

$[a) -i; b) -i; c) -1 + i; d)\ 14]$

2. Vypočtěte: [1; 19/14]

a) $[(1 + 2i) - (3 - i)](1 - i)^2$

b) $[(1 + 2i) - (3 - i)]^2(1 - i)$

c) $3(-1 + i)(1 - i) - i(2 - 3i)$

d) $\left(\sqrt{2} - i\sqrt{3}\right)i\sqrt{3} + \left(\sqrt{3} - i\sqrt{2}\right)i\sqrt{2}$

$[a) 6 + 4i; b) -17 - 7i; c) -3 + 4i; d)\ 5 + 2i\sqrt{6}]$

3. Určete reálná čísla *a, b* tak, aby platilo: [1; 19/15]

a) $(1 - i)a — (2 + i)b = 5 - 2i$

b) $\left(\sqrt{3} + i\sqrt{2}\right)a - \left(\sqrt{2} + i\sqrt{3}\right)b - 1 = i^2 - i$

$[a) a = -1; b = 3; b) a = \sqrt{2}, b = \sqrt{3}]$

4. Vypočtěte: [1; 22/ př. 3]

a) $\frac{3+4i}{2-5i}$

b) $\frac{1}{2+i}$

c) $\frac{1-i}{i}$

$\left[a) -\frac{14}{29} + \frac{23}{29}i;\ b)\ \frac{2}{5} - \frac{1}{5}i;\ c) -1 - i\right]$

5. Vypočtěte: [1; 23/ př. 4]

a) $(1 + i)^{-2}$

b) $\left(\frac{1+i}{1-i}\right)^{-1}$

$\left[a) -\frac{i}{2}; b) -i\right]$

6. Řešte v C: [1; 23/ př. 5]

$$\frac{i}{1 - 2i}x + 2i = x - \frac{1}{2 + i}$$

$\left[x = \frac{1}{10} + \frac{13}{10}i\right]$

7. Zapište v algebraickém tvaru: [1; 25/ 16]

a) $\frac{2-i}{-3+i} - \frac{1+2i}{1-3i}$

b) $2 + \frac{1}{3i} - \frac{3}{i^3}$

c) $\frac{10}{3-2i} + \frac{3-2i}{10}$

d) $\left(\frac{1-i}{1+i}\right)^{-2} + \left(\frac{1+i}{1-i}\right)^{2}$

$\left[a)\ \frac{-1}{5} - \frac{2}{5}i; b)\ 2 - \frac{10}{3}i; c)\ \frac{339}{130} + \frac{174}{130}i; d) -2\right]$

8. Řešte v C s neznámou x: [1; 25/ 19]

a) $\frac{1+2i}{1-i}x + \frac{1}{1+i} = 0$

b) $\frac{2i}{1+i}x - \frac{3-i}{2-3i} = i$

$$\left[a)\ x = \frac{2}{5} + \frac{1}{5}i; b) x = \frac{29}{26} + \frac{11}{26}\ i\right]$$

9. Určete: [1; 29/ 21]

a) $\left|1 - i + \frac{1+2i}{3-i}\right|$

b) $\big||2-3i| - i|2+i|\big|$

c) $\left|\frac{i}{\sqrt{3}+i\sqrt{2}}\right|$

d) $||2+3i|^2 + (2+3i)^2|$

$$\left[a)\ \frac{1}{10}\sqrt{130}; b)\ 3\sqrt{2}; c)\frac{1}{5}\sqrt{5}\ ; d)\ 4\sqrt{13}\right]$$

10. Řešte v C rovnice: [1; 29/ 23]

a) $|x| - x = 1 + 2i$

b) $x^2 + |x| = 0$

c) $\left(2 - \frac{1}{i}\right)\bar{x} + 2x = 10i$

d) $\bar{x}(x-1) = |x-1|^2$

$$\left[a)\ x = \frac{3}{2} - 2i; b)\ x_1 = 0;\ x_2 = i;\ x_3 = -i; c)\ x = 10 - 40i; d)\ x = 1\right]$$